RIPHAH INTERNATIONAL UNIVERSITY



Faculty of Computing

Object Oriented Programming Lab

Lab 3

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Faculty of Computing

Lab Task 1 – Student Information

Create a Student class with private name and age variables. Use setters to assign values and getters to

display them. Take name and age from the user using Scanner.

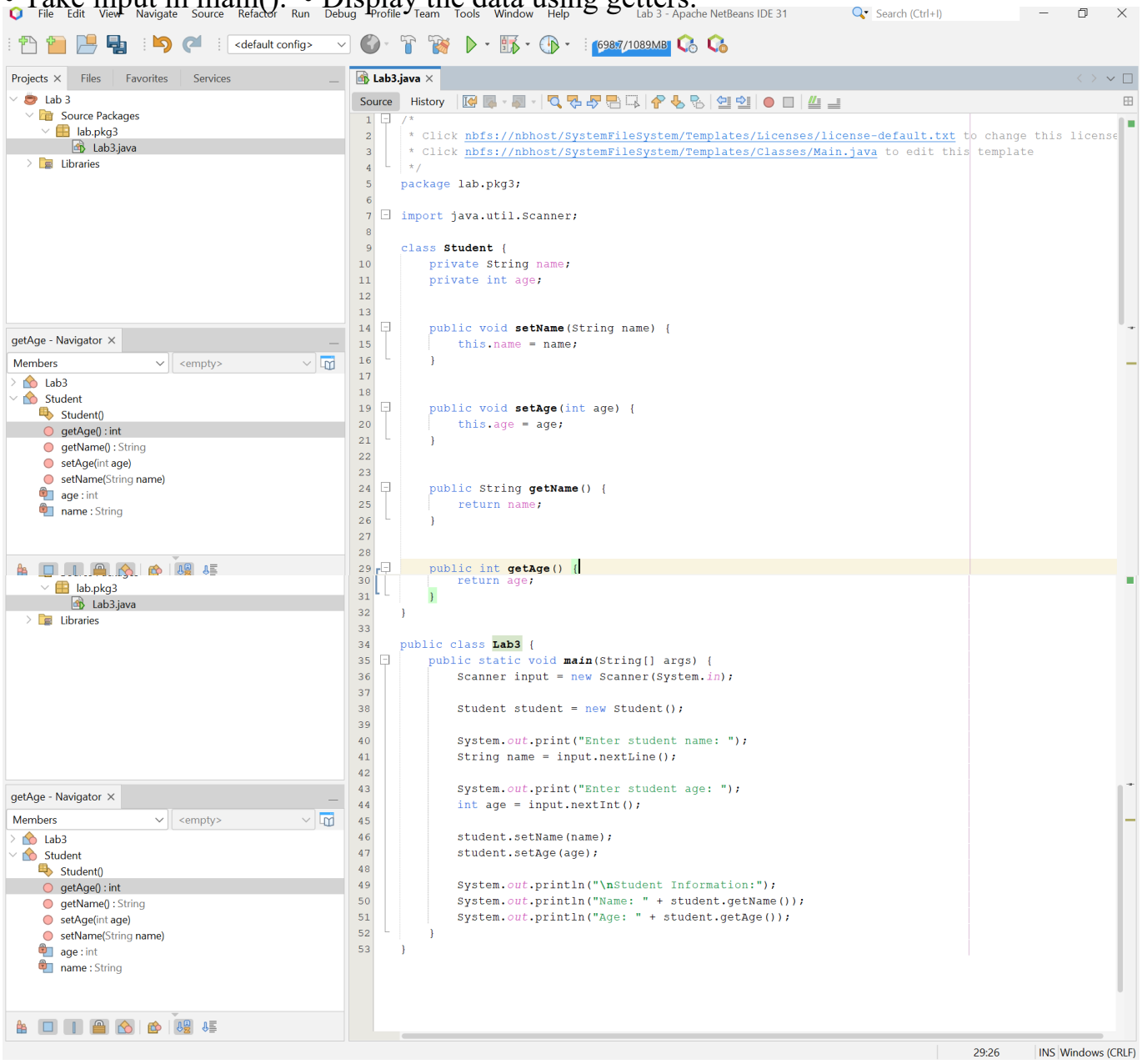
Requirements:

• Use private instance variables: name and age.

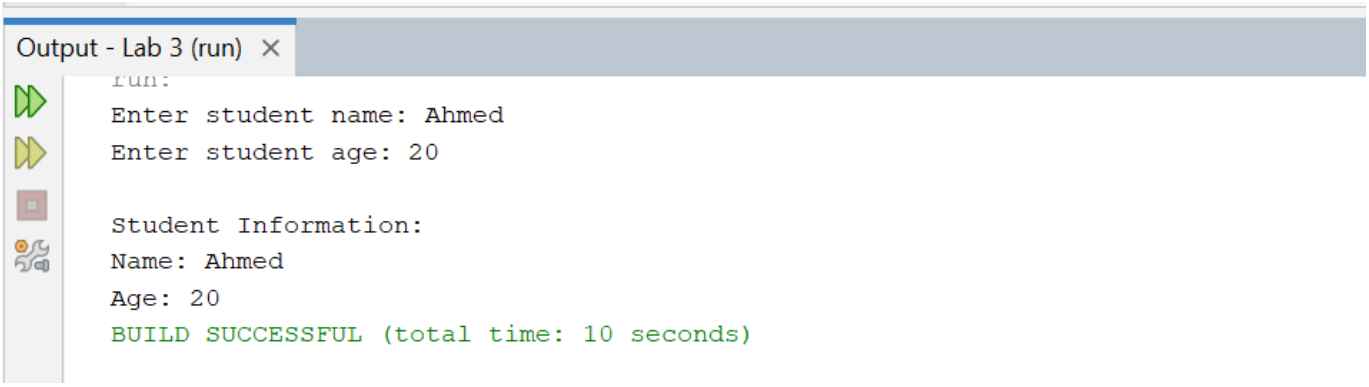
• Create setName(), setAge(), getName(), and getAge().

• Use this in the setter methods.

• Take input in main(). • Display the data using getters.



OUTPUT:



Lab Task 2 – Product Counter

Create a Product class with name, price, and quantity as private instance variables and totalProducts as

a static variable.

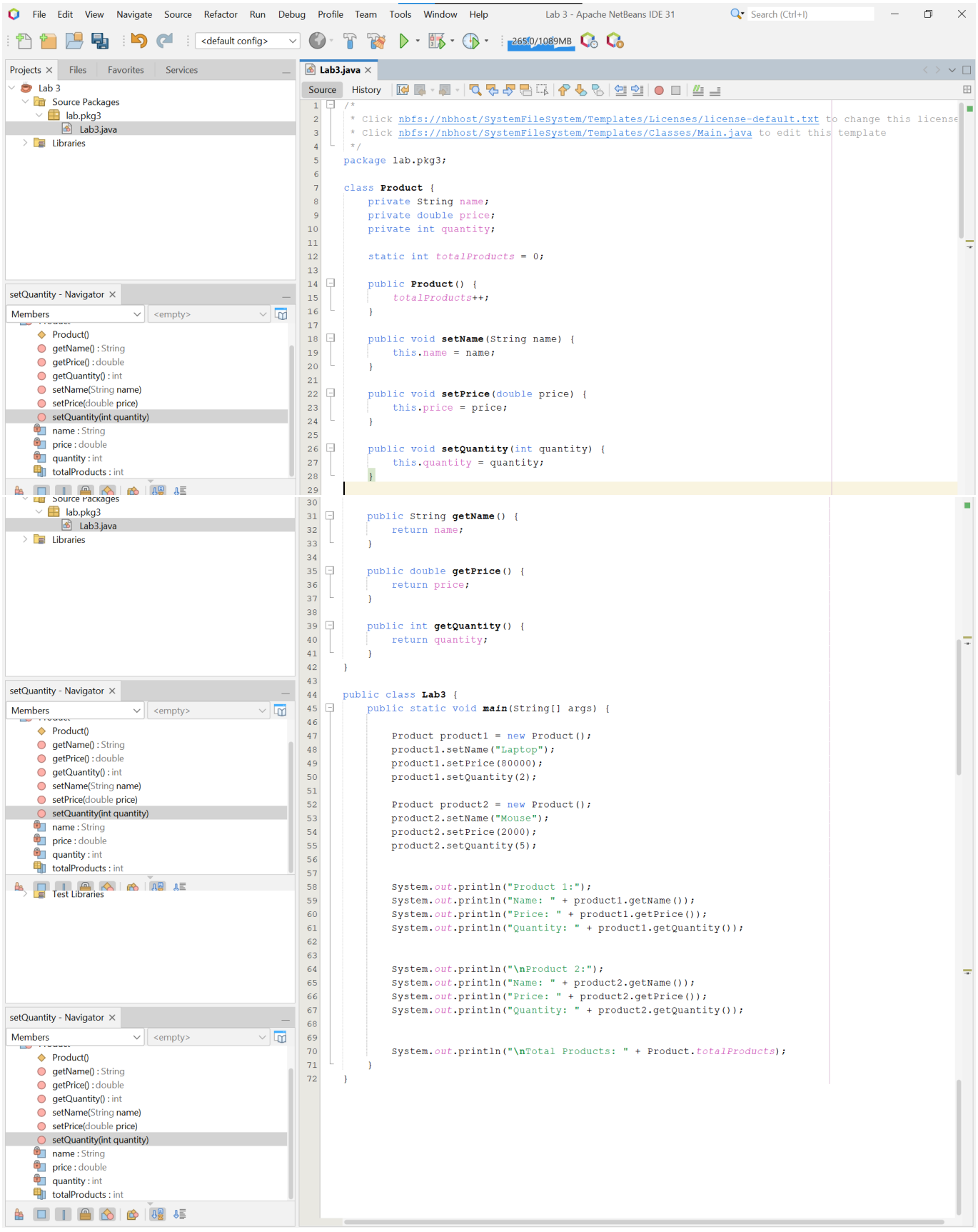
• Create setters and getters for name, price, and quantity.

• Use this in all setters.

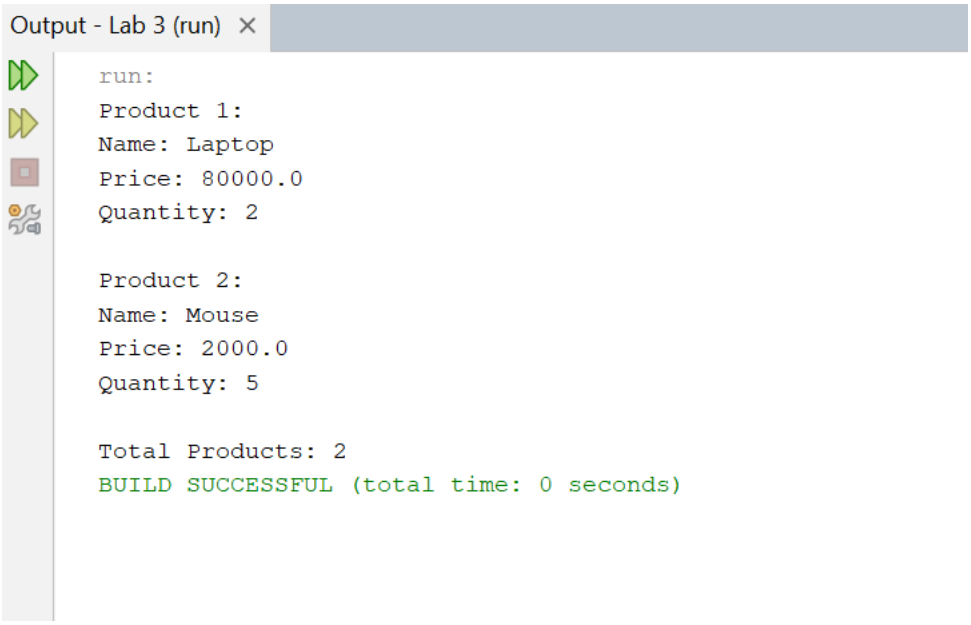
• Create two Product objects.

• Increase totalProducts whenever a product object is created or registered.

• Display both products and the total number of products.


File Edit View Navigate Source Refactor Run Debug Profile Team Tools Window Help
Lab 3 - Apache NetBeans IDE 31
Search (Ctrl+I)
<default config>
2650/1089MB
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Lab 3
Source Packages
lab.pkg3
Lab3.java
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/*
* Click nbfs://nbhost/SystemFileSystem/Templates/Licenses/license-default.txt to change this license
* Click nbfs://nbhost/SystemFileSystem/Templates/Classes/Main.java to edit this template
*/
package lab.pkg3;
class Product {
private String name;
private double price;
private int quantity;
static int totalProducts = 0;
public Product() {
totalProducts++;
}
public void setName(String name) {
this.name = name;
}
public void setPrice(double price) {
this.price = price;
}
public void setQuantity(int quantity) {
this.quantity = quantity;
}
public String getName() {
return name;
}
public double getPrice() {
return price;
}
public int getQuantity() {
return quantity;
}
}
public class Lab3 {
public static void main(String[] args) {
Product product1 = new Product();
product1.setName("Laptop");
product1.setPrice(80000);
product1.setQuantity(2);
Product product2 = new Product();
product2.setName("Mouse");
product2.setPrice(2000);
product2.setQuantity(5);
System.out.println("Product 1:");
System.out.println("Name: " + product1.getName());
System.out.println("Price: " + product1.getPrice());
System.out.println("Quantity: " + product1.getQuantity());
System.out.println("\nProduct 2:");
System.out.println("Name: " + product2.getName());
System.out.println("Price: " + product2.getPrice());
System.out.println("Quantity: " + product2.getQuantity());
System.out.println("\nTotal Products: " + Product.totalProducts);
}
}
setQuantity - Navigator
Members
<empty>
Product()
getName() : String
getPrice() : double
getQuantity() : int
setName(String name)
setPrice(double price)
setQuantity(int quantity)
name : String
price : double
quantity : int
totalProducts : int
Test Libraries

OUTPUT:



Lab Task 3 – Bank Account

Create a BankAccount class with accountHolder, accountNumber, and balance as private instance

variables. Use getters and setters. Add a static totalAccounts variable.

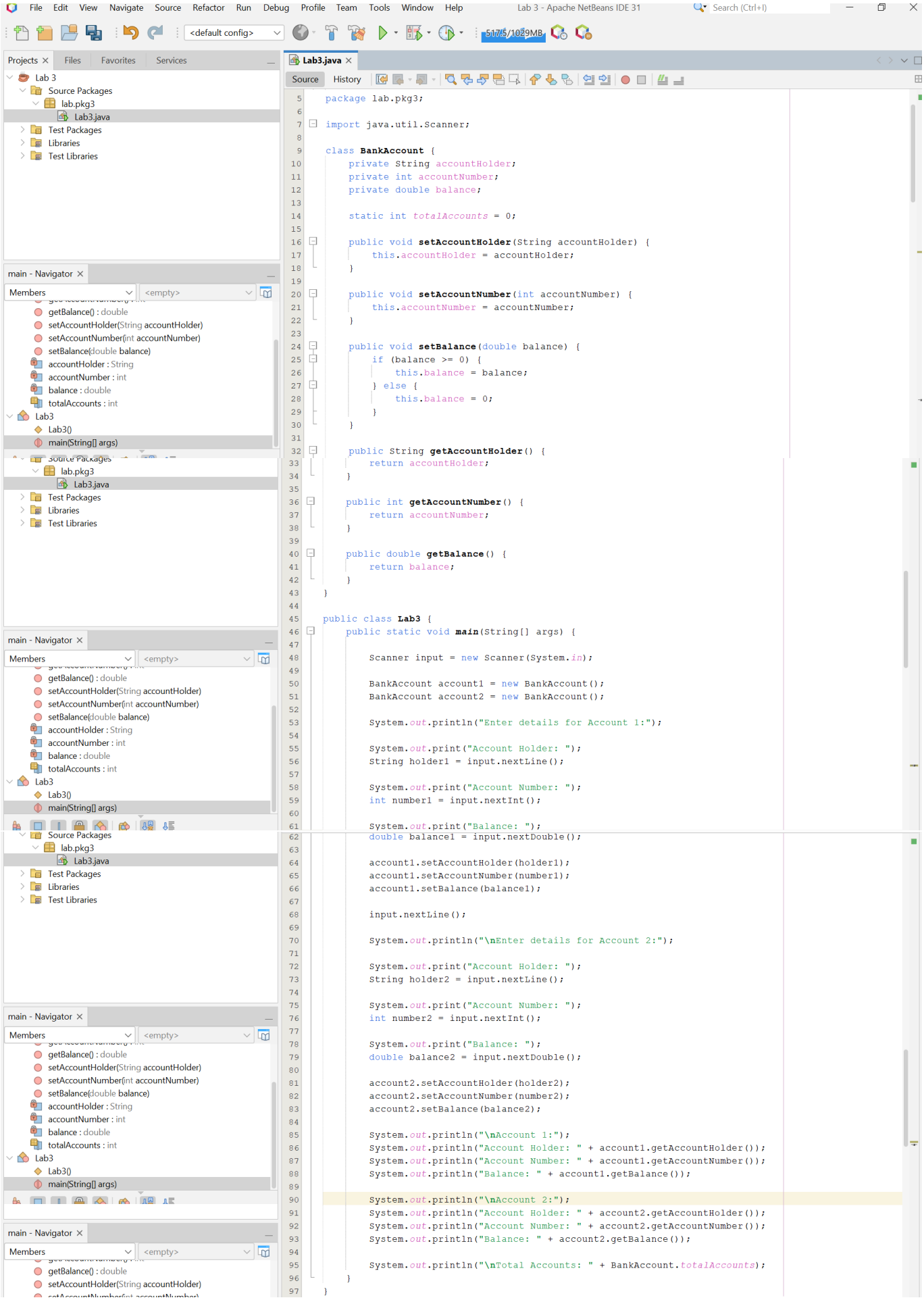
• Take input for two bank accounts using Scanner.

• Set all values through setters.

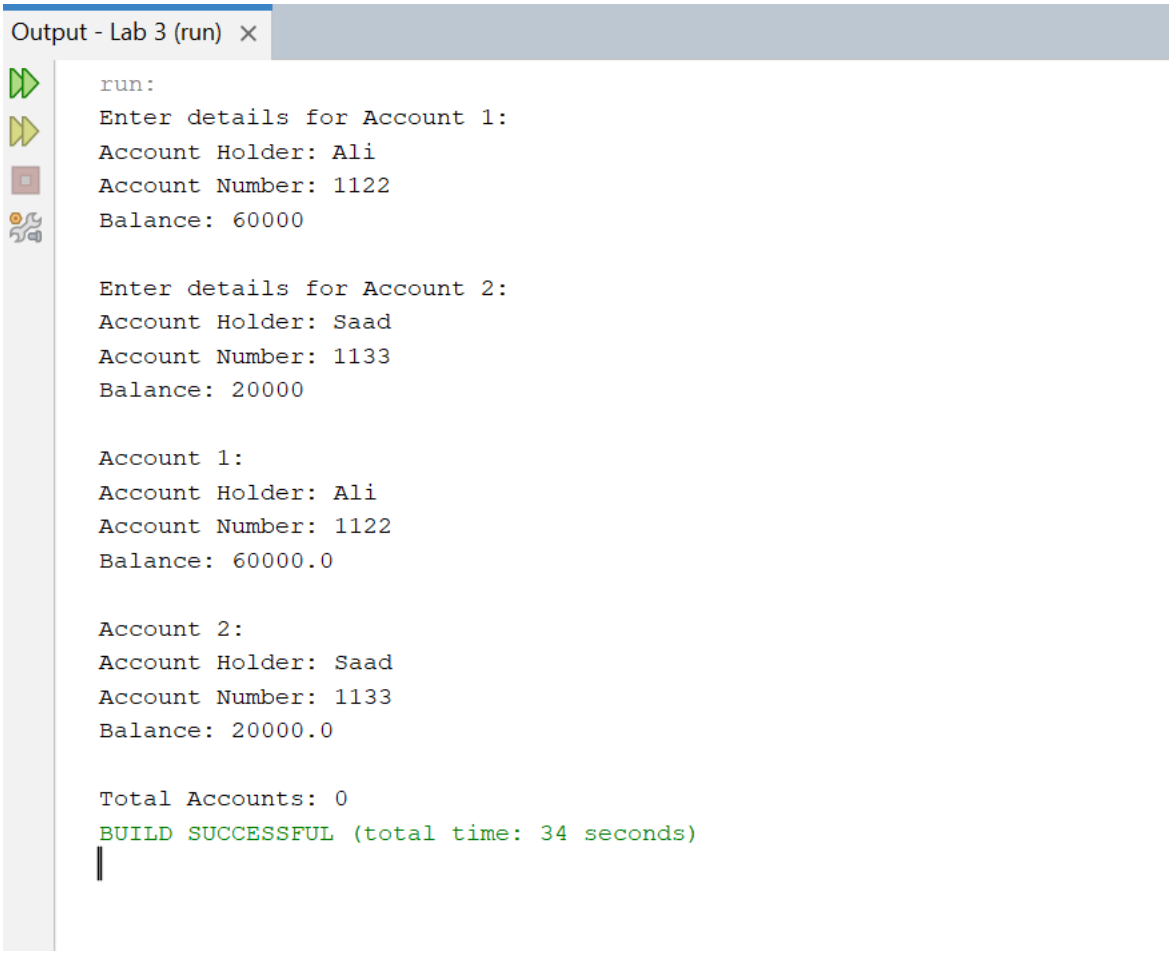
• Display values through getters.

• Use a setter to prevent a negative balance.

• Display totalAccounts using the class name.


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Lab 3
Source Packages
lab.pkg3
Lab3.java
Test Packages
Libraries
Test Libraries
main - Navigator
Members
<empty>
getBalance() : double
setAccountHolder(String accountHolder)
setAccountNumber(int accountNumber)
setBalance(double balance)
accountHolder : String
accountNumber : int
balance : double
totalAccounts : int
Lab3
Lab3()
main(String[] args)
Lab3.java
Source
History
package lab.pkg3;
import java.util.Scanner;
class BankAccount {
private String accountHolder;
private int accountNumber;
private double balance;
static int totalAccounts = 0;
public void setAccountHolder(String accountHolder) {
this.accountHolder = accountHolder;
}
public void setAccountNumber(int accountNumber) {
this.accountNumber = accountNumber;
}
public void setBalance(double balance) {
if (balance >= 0) {
this.balance = balance;
} else {
this.balance = 0;
}
}
public String getAccountHolder() {
return accountHolder;
}
public int getAccountNumber() {
return accountNumber;
}
public double getBalance() {
return balance;
}
}
public class Lab3 {
public static void main(String[] args) {
Scanner input = new Scanner(System.in);
BankAccount account1 = new BankAccount();
BankAccount account2 = new BankAccount();
System.out.println("Enter details for Account 1:");
System.out.print("Account Holder: ");
String holder1 = input.nextLine();
System.out.print("Account Number: ");
int number1 = input.nextInt();
System.out.print("Balance: ");
double balance1 = input.nextDouble();
account1.setAccountHolder(holder1);
account1.setAccountNumber(number1);
account1.setBalance(balance1);
input.nextLine();
System.out.println("\nEnter details for Account 2:");
System.out.print("Account Holder: ");
String holder2 = input.nextLine();
System.out.print("Account Number: ");
int number2 = input.nextInt();
System.out.print("Balance: ");
double balance2 = input.nextDouble();
account2.setAccountHolder(holder2);
account2.setAccountNumber(number2);
account2.setBalance(balance2);
System.out.println("\nAccount 1:");
System.out.println("Account Holder: " + account1.getAccountHolder());
System.out.println("Account Number: " + account1.getAccountNumber());
System.out.println("Balance: " + account1.getBalance());
System.out.println("\nAccount 2:");
System.out.println("Account Holder: " + account2.getAccountHolder());
System.out.println("Account Number: " + account2.getAccountNumber());
System.out.println("Balance: " + account2.getBalance());
System.out.println("\nTotal Accounts: " + BankAccount.totalAccounts);
}
}

OUTPUT:


Output - Lab 3 (run)
run:
Enter details for Account 1:
Account Holder: Ali
Account Number: 1122
Balance: 60000
Enter details for Account 2:
Account Holder: Saad
Account Number: 1133
Balance: 20000
Account 1:
Account Holder: Ali
Account Number: 1122
Balance: 60000.0
Account 2:
Account Holder: Saad
Account Number: 1133
Balance: 20000.0
Total Accounts: 0
BUILD SUCCESSFUL (total time: 34 seconds)